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GLYCOGEN HYPERACCUMULATION IN WHITE MUSCLE FIBRES OF RN⁻ CARRIER PIGS. A BIOCHEMICAL AND ULTRASTRUCTURAL STUDY

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Abstract—1. The dominant RN⁻ gene affects meat quality of pigs by increasing the glycogen content of muscle. Glycogen localization was studied in *Longissimus dorsi* muscle from RN⁻ carrier pigs (RN⁻ pigs) and rn⁺rn⁺ homozygous pigs (rn⁺ pigs).

2. Ultrastructural study showed an excess of glycogen in the sarcoplasm of white fibres from RN⁻ pigs as compared to rn⁺ pigs.

3. Lysosome-enriched fractions extracted from muscles contained 6% of the tissue glycogen content in both types of pigs. The distribution of the glycogen particles between sarcoplasm and lysosomes appeared to be similar in both RN⁻ and rn⁺ pig tissues.

4. White fibres from RN⁻ pigs with an increased glycogen level showed two ultrastructural alterations: the sarcoplasmic compartment was abnormally enlarged and a large proportion of mitochondria was morphologically modified.

5. The RN⁻ gene seems, therefore, to be associated with alterations in the glycolytic metabolism, in the distribution of proteic compartments and in the oxidative metabolism of white muscle fibres.

INTRODUCTION

A dominant gene unfavourably affecting meat quality has been recently found in a French population of pigs (*Sus scrofa domesticus*); this gene was called RN⁻, the favourable allele being rn⁺ (Naveau, 1986; Le Roy *et al.*, 1990). The RN⁻ gene affects meat quality by increasing the glycogen content of muscle, inducing the production of "acid meat" (Monin, 1989; Fernandez *et al.*, 1992). Glycogen levels are increased by about 70% in white muscles from RN⁻ carrier pigs (referred to as RN⁻), as compared with rn⁺rn⁺ homozygous pigs (referred to as rn⁺). By contrast, no effect was observed on liver glycogen content (Monin *et al.*, 1992).

When glycogen storage diseases are associated with glycogen excess, the latter can involve two localizations in muscular cells, i.e. the sarcoplasmic or the lysosomal compartment (Dreyfus, 1973; Moses, 1990). Since the biochemical mechanisms of glycogen degradation in sarcoplasm and lysosomes are different and independent, only one, as well as both, compartments could be involved (Geddes, 1986). The aim of the present study was to establish the localization of the glycogen excess in the muscle of RN⁻ pigs.

Glycogen localization was studied in *Longissimus dorsi* muscle using ultrastructural analysis and biochemical methods. RN⁻ and rn⁺ pigs were identified according to the glycolytic potential of their *Longissimus dorsi* muscle, measured from a shot biopsy. According to Fernandez *et al.* (1992), RN⁻ pigs can be discriminated on this basis: pigs with a glycolytic potential higher than 210 μmol/g can be considered as RN⁻ pigs.

MATERIALS AND METHODS

Animals

The glycolytic potential was determined on shot muscle biopsies from 25 pigs (Talmant *et al.* 1989). Glycolytic potential (an estimator of glycogen *intra vitam*) was estimated according to the following formula:

$$\text{Glycolytic potential} = 2([\text{glycogen}] + [\text{glucose-6-phosphate}] + [\text{glucose}]) + [\text{lactic acid}].$$

This equation takes into account the glycogenolysis occurring during taking and freezing biopsied samples (Monin *et al.*, 1985). Four RN⁻ pigs with glycolytic potentials between 299 and 328 μmol/g fresh muscle and 4 rn⁺ pigs with glycolytic potentials between 161 and 179 μmol/g fresh muscle, were retained for the study. The animals were slaughtered at around 100 kg live weight by electronarcosis and exsanguination in an experimental slaughterhouse, in four slaughter series. Each series included 1 RN⁻ pig and 1 rn⁺ pig. Around 100 g of *Longissimus dorsi* were taken for biochemical study. Just before slaughter, a muscle sample was taken by shot biopsy for microscopic study.

Preparation of a lysosome-enriched fraction

Mixed lysosomal/mitochondrial fractions (L + M) were prepared by the method of Geddes and Stratton

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Table 1. Distribution of *N*-acetyl-β-D-glucosaminidase activity in fractions obtained by differential centrifugation of a skeletal muscle homogenate

	RN⁻ (pkat/mg protein)	rn⁺ (pkat/mg protein)	RN⁻ (% of total activity)	rn⁺ (% of total activity)
H1	8.0 ± 1.6	8 ± 1.1	100	100
H2	5.6 ± 0.8	5.9 ± 0.8	24 ± 5	23 ± 2
H3	1.0 ± 0.1	1.5 ± 0.1	2.0 ± 0.5	0.6 ± 0.1
S4	4.0 ± 0.3	4.0 ± 0.6	23 ± 2	24 ± 1
(L + M)	177 ± 21	158 ± 32	32 ± 3	27 ± 1

No significant difference was observed between RN⁻ ($N = 4$) and rn⁺ ($N = 4$) pigs. Values are means ± SEM.

(1977) slightly modified. One hundred grams of muscle were taken, quickly cut into cubes of approximately 1 cm³, and immediately immersed in 1 l of cold homogenizing buffer (150 mM KCl, 2 mM EDTA, 20 mM Tris, 5 mM iodoacetate, pH 7.5, 10 ml/g tissue). The mixture was homogenized with a Waring Blender (2 × 20 sec). The resulting homogenate (H1) was centrifuged at 1000 *g* for 10 min at 4°C. The pellet, containing heavy organelles and cell debris, was again homogenized in the same buffer with a Waring Blender (2 × 20 sec) (H2). The supernatant was filtered through eight layers of cheesecloth, and recentrifuged at 3000 *g* for 5 min at 4°C. The pellet was suspended in homogenizing buffer (H3) with a Waring Blender (2 × 20 sec).

The supernatant was centrifuged at 20,000 *g* for 20 min at 4°C in order to separate the supernatant (S4) from the lysosomal/mitochondrial fraction (L + M). This fraction was gently resuspended in homogenizing buffer using a Potter–Elvehjem. Each fraction, H1, H2, H3, S4, (L + M) was immediately collected at each step and divided into several parts which were frozen at −20°C for further determinations of *N*-acetyl-β-D-glucosaminidase activity, glycogen and protein contents. Prior to freezing, the parts taken for glycogen determination were diluted in perchloric acid at a final concentration of 0.5 M.

Analytical techniques

N-Acetyl-β-D-glucosaminidase activity (Caroll, 1978). *N*-Acetyl-β-D-glucosaminidase was chosen as marker enzyme for lysosomes and used to estimate the yield in lysosomes. Samples (100 μl) were incubated with 0.5 ml of 5 mM 4-nitrophenyl *N*-acetyl-β-D-glucosaminidase in 0.05 M sodium citrate buffer, pH 4.4. After 30 min at 37°C, the reaction was stopped by adding 2.5 ml of 0.5 M glycine NaOH buffer, pH 10.4. The liberated aglycone was measured at 420 nm.

Table 2. Distribution of protein in fractions

	RN⁻ (mg/g tissue)	rn⁺ (mg/g tissue)	RN⁻ (%)	rn⁺ (%)
H1	109 ± 6	127 ± 3	100	100
H2	36 ± 3	35 ± 2	33 ± 2	28 ± 2
H3	0.7 ± 0.2	1.0 ± 0.4	0.6 ± 0.2	0.8 ± 0.3
S4	51 ± 5	56 ± 6	47 ± 5	44 ± 4
(L + M)	1.5 ± 0.1	2.1 ± 0.5	1.8 ± 0.1	1.1 ± 0.4

No significant difference was observed between RN⁻ ($N = 4$) and rn⁺ ($N = 4$) pigs. Results are means ± SEM.

Table 3. Distribution of glycogen in fractions

	RN⁻ (μmol/g tissue)	rn⁺ (μmol/g tissue)	RN⁻ (%)	rn⁺ (%)
H1	92 ± 3**	53 ± 6	100	100
H2	14 ± 1*	9 ± 2	15 ± 2	16 ± 2
H3	0.6 ± 0.1	0.4 ± 0.1	0.6 ± 0.1	0.9 ± 0.2
S3	73 ± 3**	41 ± 4	79 ± 2	77 ± 2
L + M	2.0 ± 0.1**	1.4 ± 0.2	2.2 ± 0.1	2.5 ± 0.2

Results are means ± SEM. Significant differences between RN⁻ and rn⁺ pigs. *$P < 0.05$, **$P < 0.01$.

Protein determination. The protein content was assayed by the method of Lowry *et al.* (1951), using bovine serum albumin (BSA) as standard.

Glycogen determination. Aliquots of 0.5 ml were taken for the enzymatic determination of glycogen hydrolysis by amyloglucosidase (Dalrymple and Hamm, 1973).

Data were represented as mean ± SEM. The mean values were compared by Student's *t*-test.

Electron microscopy

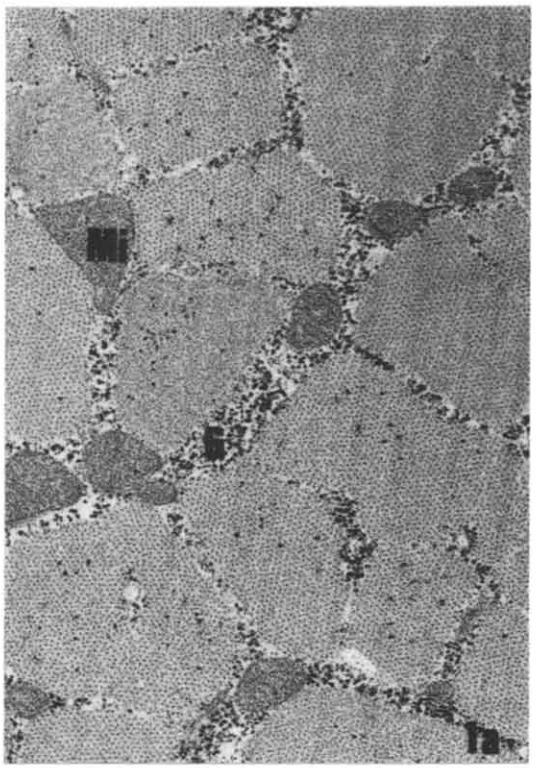
Small blocks (1–2 mm³) were cut from the muscle biopsies, fixed for 2 hr at 4°C in 2.5% glutaraldehyde in 0.1 M cacodylate buffer pH 7.0 and post-fixed in 1% osmium tetroxide in the same buffer for 1 hr at 4°C. The specimens were dehydrated through an ethanol gradient and embedded in epoxy resin. Glycogen was stained on ultrathin sections by the periodic acid–thiocarbohydrazide (TCH)–silver proteinate method (PATAg) according to Thiéry (1967). The specificity of the staining reaction for polysaccharides was evaluated by omitting the oxidation by periodic acid in control sections. The observations were made with a Philips EM 400 electron microscope at an accelerating voltage of 80 kV.

RESULTS

Biochemical study

Table 1 shows *N*-acetyl-β-D-glucosaminidase activity at each step during the preparation of the (L + M) fractions. The specific activity was similar in the whole homogenates (H1) from both RN⁻ and rn⁺ pigs (around 8 pkat/mg protein). During the purification process, some lysosomes were lost or ruptured, since all intermediate fractions contained the enzyme activity. The final pellet contained approximately 30% of total lysosomes. The loss occurred in the same proportion in the tissues from RN⁻ and rn⁺ pigs. Enrichment factors in lysosomes were identical, i.e. about 20%, so it was possible to compare the lysosome-enriched fractions from both RN⁻ and rn⁺ pigs. *N*-Acetyl-β-D-glucosaminidase activities in the final pellets were 177 and 158 pkat/mg protein for RN⁻ and rn⁺ pigs, respectively. These values were not significantly different.

Table 2 shows the distribution of protein in the fractions. Protein content was lower in muscle from RN⁻ pigs as compared to rn⁺ pigs. This difference



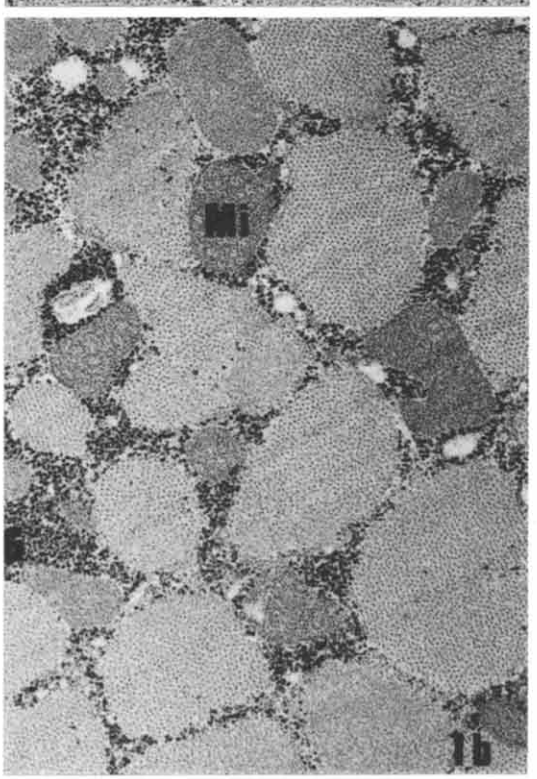
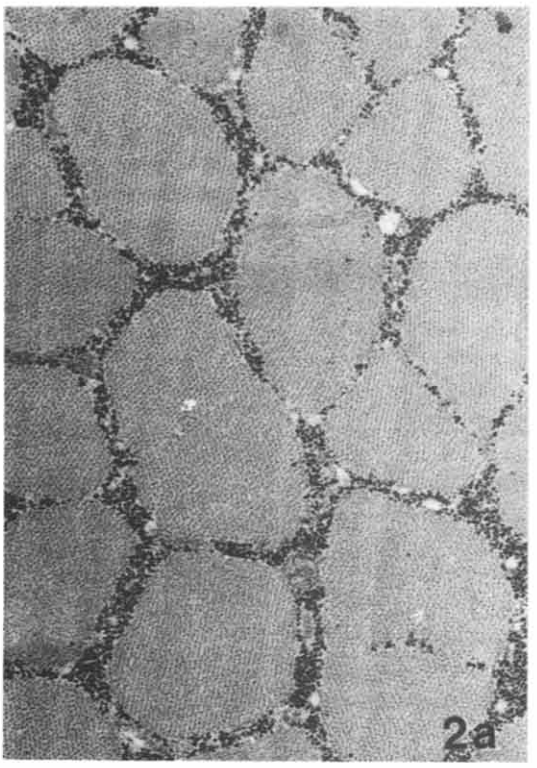


Fig. 1. Electron micrographs of PATAg-stained transverse sections of oxidative fibres from *Longissimus dorsi* biopsies. 1a: muscle of rn^+ pig; 1b: muscle of RN^- pig. G: glycogen; Mi: mitochondria (×16000).



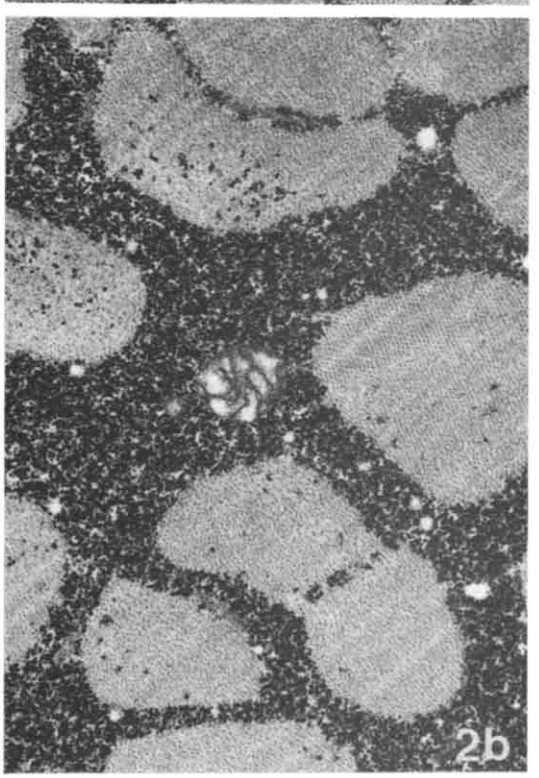
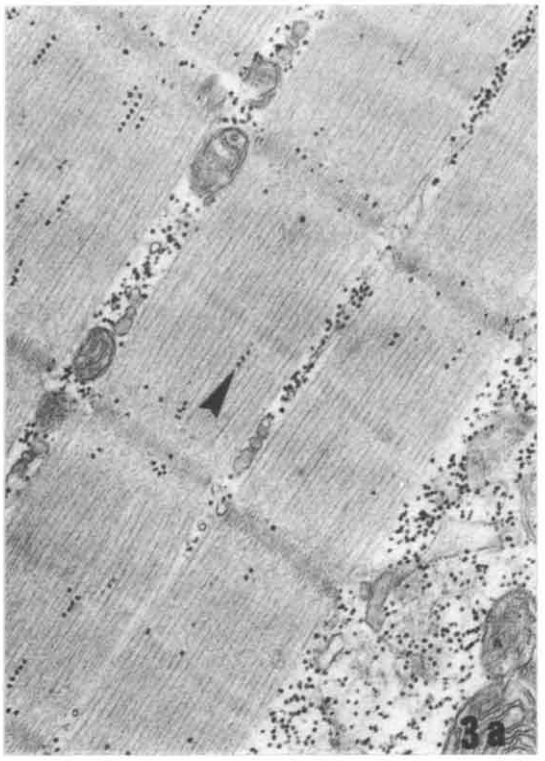


Fig. 2. Transverse sections of glycolytic fibres from *Longissimus dorsi* biopsies. 2a: muscle of rn^+ pig; 2b: muscle of RN^- pig. Note the difference in the size of the sarcoplasmic compartment between rn^+ and RN^- muscle (×16000).



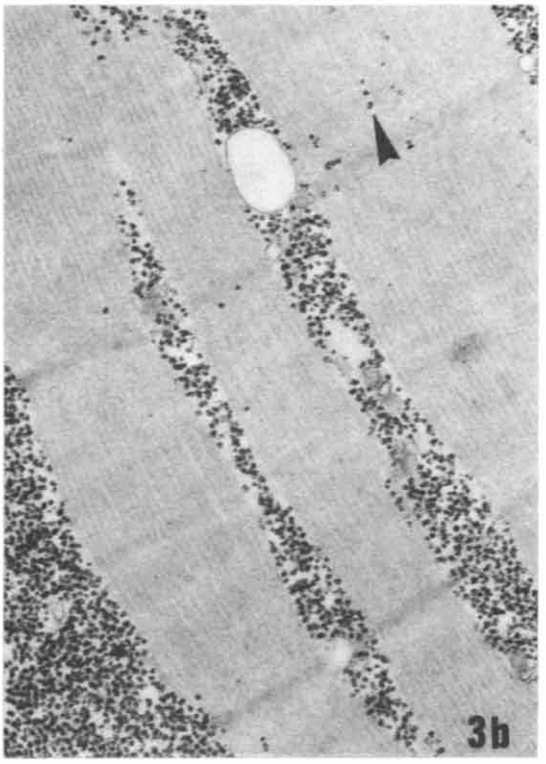
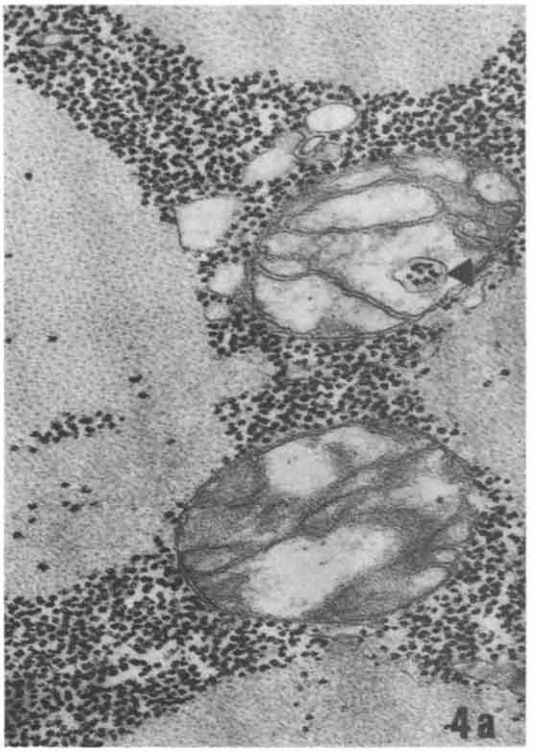


Fig. 3. Longitudinal sections of glycolytic fibres from *Longissimus dorsi* biopsies. 3a: muscle of rn^+ pig; 3b: muscle of RN^- pig. Glycogen particle localization was mainly intermyofibrillar and subsarcolemmal. Occasionally, particles were seen between the myofilaments inside myofibrils (arrows) (× 28000).



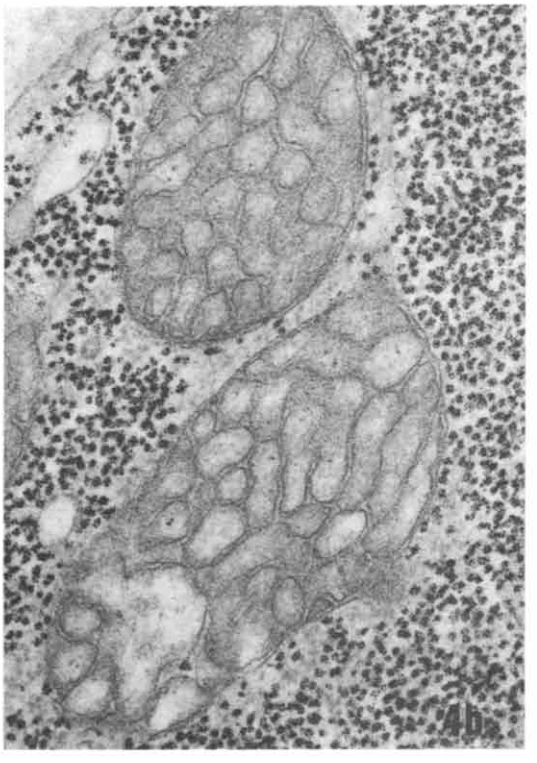


Fig. 4. Abnormal mitochondria in sections from *Longissimus dorsi* biopsies of RN^- pigs. 4a: glycogen particles are seen in the sarcoplasm and within a mitochondria (arrow). 4b: mitochondrial cristae are irregularly shaped (× 54000).

was kept in H2, S4 and (L + M) fractions. Approximately 1.5% of the total protein was contained in the (L + M) fraction in both types of pigs.

Table 3 shows the distribution of glycogen in the fractions. Glycogen was present in all fractions. The glycogen level in the total muscle homogenate was 70% higher in RN⁻ pigs than in rn⁺ pigs. This excess was kept in all the fractions. The final lysosome-enriched pellet contained some glycogen, i.e. 2 and 1.4 μmol/g tissue for RN⁻ and rn⁺ pigs, respectively. The distribution of glycogen in fractions was similar in RN⁻ and rn⁺ tissues: almost 80% of total glycogen were found in the S4 fraction and about 2.5% in the (L + M) fraction.

Electron microscopy

After PATAg staining, glycogen appeared as distinct particles in the sections. The size of these particles varied from 10 to 40 nm. Particles were found in all fibres. Their localization was intracellular, mainly in subsarcolemmal and intermyofibrillar spaces. Most of the glycogen appeared as free particles in the sarcoplasm. Occasionally, these particles were packed in lysosomes or seemed to be linked intimately with the myofilaments.

Based on the mitochondrial content, individual fibres were classified into oxidative fibres and glycolytic fibres (Eisenberg, 1976). The oxidative fibres exhibited a low content of glycogen. Glycogen localization and ultrastructural pattern were similar in RN⁻ and rn⁺ pigs. However, a higher density of glycogen particles was observed in fibres from RN⁻ pigs (Fig. 1). Intramyofibrillar and lysosomal glycogen were occasionally observed. In glycolytic fibres, glycogen particles were abundant. They were located in all the sarcoplasm. The arrangement of particles was homogeneous in fibres from both RN⁻ and rn⁺ pigs. A much higher amount of glycogen was observed in fibres of RN⁻ pigs as compared with those of rn⁺ pigs. In glycolytic fibres from RN⁻ pigs, large deposits of subsarcolemmal glycogen particles were frequently observed, and the intermyofibrillar spaces were broad and irregular. Conversely, in fibres of rn⁺ pigs, subsarcolemmal deposits were occasionally seen, too, but to a much lesser extent, and the intermyofibrillar spaces were regular and narrow (Fig. 2).

Particles of glycogen were also frequently seen inside myofibrils in both types of animals. These clusters consisted of single rows of particles orientated longitudinally between the thin filaments, close to the Z-disc and, in some cases, in the A-band. The number of intramyofibrillar particles was low, seeming slightly higher in fibres of RN⁻ pigs than in those of rn⁺ pigs (Fig. 3).

A small proportion of glycogen particles was enclosed in lysosomes. The size and the number of lysosomes containing glycogen varied between fibres and between animals.

Ultrastructural abnormalities were observed in some mitochondria from RN⁻ pigs and consisted of disorganised cristae or glycogen particles in enlarged intracristal spaces (Fig. 4). These abnormalities were specific to glycolytic fibres.

DISCUSSION

As noted by Calder and Geddes (1989), the fractionation of tissue homogenates by differential centrifugation has the advantages of speed and ease, but the disadvantage of damage to the organelles to some extent. The lysosome-enriched fraction contained approximately 30% of the total tissue activity of *N*-acetyl-β-D-glucosaminidase. This proportion was lower than that found in a homologous fraction from rat skeletal muscle by Calder and Geddes (1989). These authors found 45–50% of the total lysosomal enzyme activity in the (L + M) fraction. This difference could be explained by differences in species and in the technique of homogenization. Indeed, there may exist a higher fragility of these organelles in pig muscles as compared to rat muscles.

The lower level of protein in RN⁻ pigs as compared to rn⁺ pigs was previously reported by Monin (1989), although the difference between the two types of pigs is higher in this study.

The final pellet contained glycogen. Muscle lysosomes usually contain a small fraction of the total glycogen, which is hydrolysed to glucose by lysosomal acid-α-glucosidase and recycled to cytosol (Calder and Geddes, 1989). In the present study, the (L + M) fractions from both RN⁻ and rn⁺ tissues contained about 2% of total glycogen. Glycogen in this fraction is mainly provided by lysosomes, since the number of glycogen particles in mitochondria observed by microscopy was very low. It must be considered that this lysosomal glycogen is probably underestimated because of the loss of lysosomes during the preparation procedure. As only about 30% of the lysosomes were kept in the (L + M) fraction, it can be assumed that about 6% of the total glycogen was effectively localized in the lysosomes. Geddes and Calder (1989) found 4% of the total glycogen and 40–50% of the total lysosomal enzyme activity in their (L + M) fraction. Applying the same calculation to their results, this would indicate that the proportion of muscle glycogen contained in lysosomes of the rat would be approximately 8–10%. Muscle from pigs seems, therefore, to contain a lower proportion of lysosomal glycogen as compared to rat muscle. A second cause of underestimating the lysosomal glycogen could be the *post mortem* differential degradation of glycogen between sarcoplasm and lysosomes. Glycogenolysis in sarcoplasm was inhibited by adding iodoacetate in homogenizing buffer. This was not the case for lysosomal glycogen. Part of the latter was possibly degraded, leading to a lower proportion at the end of the preparation process. The proportion of lysosomal glycogen, with regard to total glycogen, remained normal in RN⁻ pigs. It seems, therefore, that the function of glycogen

storage by lysosomes is normal in RN$^-$ muscle. *N*-Acetyl-β-D-glucosaminidase activity in (L + M) fractions was similar in both RN$^-$ and rn$^+$ tissues. This result suggests that the glycogen increase in lysosomal fraction was not linked to an increase in lysosome content but to an increase in the concentration of the glycogen inside the lysosomes. The lysosomal compartment accumulates cytosolic material by autophagy (Geddes and Stratton, 1977). It can be assumed that this process is proportional to the sarcoplasmic glycogen quantity, which would explain the identical proportions between glycogen concentrations in the sarcoplasm and in the lysosomal compartment in both types of pigs.

The difference in glycogen level in *Longissimus dorsi* muscle between RN$^-$ and rn$^+$ pigs was mainly observed in glycolytic fibres. This observation is in good agreement with those of Monin *et al.* (1987), who reported that the excess of muscle glycogen is closely and positively correlated with the glycolytic metabolism intensity of muscle in Hampshire crossbred pigs, which can be considered as RN$^-$ pigs. Hence, the expression of the RN$^-$ gene on glycolytic potential level appears to depend on the metabolic type of muscle fibre.

Glycogen localization was mainly sarcoplasmic, a small proportion of glycogen being found in lysosomes or inside the myofibrils. The distribution of the glycogen particles appeared to be similar in both tissues from RN$^-$ and rn$^+$ pigs, which agrees with the biochemical results. Nevertheless, in the glycolytic muscle fibres of RN$^-$ pigs, the amounts and the densities of the glycogen particles were higher than in the glycolytic muscle fibres of rn$^+$ pigs. Moreover, these fibres, in the case of RN$^-$ pig muscle, showed an abnormally large sarcoplasmic compartment which was broad, irregular in shape, and the diameter of the myofibrils was very variable. As the total content of sarcoplasmic glycogen is the product of the concentration by volume of the sarcoplasmic compartment, two hypotheses can be put forward: (1) the primary defect would be an increase in the volume of the sarcoplasmic compartment, with a normal concentration of glycogen; this would indicate a deficiency in the proportions of the different protein compartments (myofibrillar, sarcoplasmic, connective); and (2) the excess of glycogen would result primarily from an increase in the concentration of the glucid in the sarcoplasm, leading to an increase in the volume of this compartment; this would suggest that the RN$^-$ gene is associated with an abnormality in the glycogen metabolic pathway, e.g. excess of synthesis, default of catabolism or defect in the mechanisms regulating the glycogen concentration. This metabolic defect could be in relation to the observed mitochondrial abnormalities. Some mitochondria showed either disorganised cristae or glycogen particles in enlarged intracristal spaces. Such abnormalities have already been observed in some myopathies (Coquet, 1991). Since mitochondria are the support of oxidative metabolism in muscle, there may be a relationship in RN$^-$ glycolytic fibres between the high glycolytic potential and a disfunction of a part or of the totality of mitochondria. However, the fact that modified mitochondria were only observed in glycolytic fibres but not in oxidative fibres of the same muscle could, in turn, indicate that the default in mitochondria is only a consequence of the other modifications observed in this type of fibre.

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